Subrat Rout

+971 526570569 | subrat258@gmail.com | https://www.linkedin.com/in/subrat-routdatascientist/ | Notice Period - 15 days

SUMMARY

AI & Machine Learning Engineer with 12+ years of experience in building AI systems and **ML solutions** across industries like fintech, logistics, healthcare, e-commerce, and media. I have hands on experience with **LLMs, RAG pipelines** and **Computer Vision Models** and worked across the full lifecycle from data gathering and modeling to deployment and integration. Over the years, I have built and scaled AI-driven products to **400K+** users, including solutions like **predictive and marketing models**. I am experienced in bridging business strategy with practical AI implementation driving revenue growth, reducing marketing spend, and improving customer retention.

KEY HIGHLIGHT

- **Zero-to-One Builder**: Launched 2 successful business platforms and AI-powered products in MENA region and scaled to 400K+ users with $2M+ MRR.
- **Gen AI & Voice Agents:** Worked on building **Voice AI agents and LLM-based systems (RAG, LoRA/PEFT)**, improving automation, accuracy, and real-time decision-making.
- **AI-Driven Business Impact:** Designed AI and ML solutions for **fraud detection, customer segmentation**, and revenue optimization, improving conversion, retention, and operational efficiency.

SKILLS

- **Generative AI & LLMs**: LangGraph, LangChain, Multi-Agent Systems, RAG, LLM Fine-tuning (PEFT, LoRA), Azure Open AI, Azure Foundry, Azure AI Search
- **Machine Learning & Data Science**:
  - Algorithms: Classification, Regression, Clustering (K-Means), Random Forest, XGBoost, Fraud Detection Models, Computer Vision Models
  - Libraries: Hugging Face Transformers, Pandas, Scikit-learn, matplotlib, numpy, OpenCV.
  - Methodologies: Predictive Modeling, User Segmentation, Funnel Analysis, MMM, MLOps, Model Drift Monitoring.
- **Data Engineering & Cloud Platforms**:
  - Cloud Databases/Warehouses: Azure Data Bricks, ADLS, Azure Synapse, AWS Cloud, Google Cloud, SQL Server, MySQL, PostgreSQL.
  - ETL/Data Processing: Data Modeling, Data Warehousing, Data Collection.
- **Business Intelligence & Analytics**:
  - Tools: **Python,** Git, SQL, CI/CD, Power BI (DAX, Power Query/M Language), Tableau, Looker Studio, Microsoft BI Stack, Microsoft Fabric
  - Web Analytics: Google Analytics, Firebase, Mo Engage

WORK EXPERIENCE

Nyukt.AI - (Contract) Data Science Manager — **Dubai** September 2025 – Present

- Worked on the development of a **conversational Design** & human-like multilingual **AI voice system** capable of handling more than 1000+ calls daily. The platform integrates speech-to-text (STT), text-to-speech (TTS), and LLM to deliver natural interactions.
- Reduced manual appointment scheduling and admin work by **70%** by building a **multilingual AI voice system** that autonomously manages registration, rescheduling, and cancellations, without human intervention.
- Used multimodal AI and prompt engineering to build reliable, real-time systems, achieving **95% call completion** rate with accurate and consistent responses.
- Worked on **RAG pipelines** utilizing **Azure OpenAI and Azure AI Search**, implemented embedding, vector search, and hybrid retrieval strategies.
- **Domain-Specific LLM Optimization**: Optimized clinically-grade LLMs using **PEFT (LoRA)** on proprietary healthcare datasets, achieving a **15% higher accuracy** in medical suggestion compared to base foundation models (OpenAI, Claude).
- Improved model reliability by building an evaluation framework with checks and rollback mechanisms to handle model drift.
- Contributed to MLOps practices including CI/CD pipelines, model & prompt versioning, using version control systems like Git.

Robustrade DMCC - (Contract) AI & Machine Learning Engineer — **Dubai** October 2024 – August 2025

- Built the analytics infrastructure for a **fintech**(mobile money solution) platform in Africa, supporting its scale to **400K+** users and MRR of 2M USD.
- Worked on developing a **Computer Vision model**–based facial recognition solution using MTCNN and FaceNet for real-time detection and high-accuracy identity verification, enabling secure onboarding of the users.
- Worked on the development of **AI-driven KYC agent** using **LangGraph** and **Azure Document Intelligence** with **Computer Vision**(**OpenCV**), automating user onboarding and reducing manual processing time by **60%** & maintaining regulatory compliance.

- Designed ML-driven **customer segmentation** and CLV model, using marketing mix modelling **(MMM)** insights to optimize acquisition and retention strategies, resulting in a **15%** increase in retention and a **20%** reduction in **CAC**.
- Developed **Azure-based** data pipelines integrated with **MS Fabric**, implementing star schema and created self-service tables to support multi-country, multi-timezone, and multi-currency analytics.
- Utilized Python for data analysis and **Power BI data** modeling and DAX optimization, significantly reducing manual effort by 50% and enhancing delivery of reports.
- Designed **CRM-driven customer segmentation** and profiling models to identify high, mid, and low value cohorts, enabling targeted and data-driven marketing campaigns.

TruKKer (MoXey Fintech) - (UAE & KSA) AI & Data Analytics Manager **Dubai** October 2022 – September 2024

- Built the analytics infrastructure for a **fintech platform** in UAE & KSA, supporting its scale to ARR of **100M AED**.
- Designed and **ML-based prediciton model** (XGBoost, Random Forest), reducing fraud cases by **20%** across Middle East markets.
- Built **K-Means** clustering model to identify high-value customer segments from transactional data, enabling more effective targeted marketing strategies.
- Developed **RFM models** and performed customer profiling based on transactional behavior & App engagement.
- Built a social media **sentiment analysis** dashboard based on social data about the company using NLP.
- Drove analytical adoption and self-service reporting by standardizing **70+ reports into 25+ streamlined** executive dashboards, utilizing reusable semantic models and visualization best practices
- Built **data pipeline** with golden dataset design, star schema, and MS Fabric data flows, reducing data inaccuracies b**y 80%.**
- Built a robust Data Warehouse using Azure Databricks and SQL, establishing a single **source of truth** across different business units in Africa and the UAE.
- Led the implementation of structured data models (**Star Schema**) for key data pipelines, improving data accuracy, reducing duplication, and enhancing data processing efficiency for reporting and analytics.

ASIANET NEWS NETWORK – Data & Growth Analytics Lead **India** May 2017 - September 2022

- Managed user and traffic growth for all 7 channels at Asianet News, achieving over 1B+ Page Views (PVs) and 90M+ Monthly Active Users (MAUs).
- Built unique audience profiles using Audience Manager for sales teams to identify their most valuable segments and utilize them across any digital channel.
- Worked on K-Means model, which was used to create clusters & generate segmented profiles of customers. The clusters Information was used in FB marketing to lower the Cost per Acquisition (CPC)
- Have set up the entire reporting structure framework for all the channels (from the analytics report to content report).
- Building unique audience profiles using Audience Manager for sales so they can identify their most valuable segments and use them across any digital channel.

HINDUSTAN TIMES - Strategy & Ops **India** July 2016 – April 2017

- Reported directly to the CEO, collaborating with product owners & stakeholders to expand the FM Radio category across PAN India.
- Provided periodic MIS reporting to the CEO and prepared performance presentations. Interfaced with selected clients, managed budgeting, and handled special situations.

INDIAMART - Category Manager **India** May 2015 - Jul 2016

- Worked directly with senior management and key stakeholders, including the CEO, to expand the Business of the Tool category unit of IndiaMart into the B2B market, with an emphasis on aggressive expansion into the SME segment.
- Led the creation of a 6-month growth business roadmap to deliver a GMV of 3.2 crore, utilizing business demographics and forecasting models, and executed plans based on consensus among key management and division heads.

ADITYA BIRLA GROUP Senior Engineer **India** July 2011 - Apr 2014

- As part of the procurement team, developed the concept of an Integrated Vendor Experience Centre, which reduced merchant onboarding complexities, centralized ordering, and improved auditing.
- Handled the onboarding of over 200+ vendors within two months for material supply.

EDUCATION

MBA, Operations and Strategy | Great Lakes Institute of Management | 2015 | 3/4 (Deans List)

Bachelor of Engineering |VSSUT, Burla – India | 2011 | 80%